

EXP.NO: 3	Shell Scripting
DATE: 30/01/2026	

AIM:

To write a Shell script to display Fibonacci Series.

ALGORITHM:

1. **Start**
2. **Input:** Get the number of terms (\$n\$) from the user.
3. **Initialize:** * Set the first number (\$a\$) to 0.
 - Set the second number (\$b\$) to 1.
4. **Loop:** Start a counter (\$i\$) from 0 and repeat until $i < n$:
 - **Print:** Display the current value of \$a\$.
 - **Calculate:** Use expr to find the next term: $\$next = a + b$.
 - **Update:** * Move the value of \$b\$ into \$a\$ ($a = b$).
 - Move the value of \$next\$ into \$b\$ ($b = next$).
 - **Increment:** Add 1 to the counter ($i = i + 1$).
5. **End**

CODE:

```
#!/bin/bash
# Fibonacci Series using expr

echo "Enter the number of terms:"
read n

a=0
b=1

echo "The Fibonacci series is:"

for (( i=0; i<n; i++ ))
do
    echo -n "$a "
    # Using expr for addition
    fn=$(expr $a + $b)
    a=$b
    b=$fn
done
```

```
done  
echo ""
```

OUTPUT:

```
Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System  
$ vi fib.sh  
  
Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System  
$ chmod +x fib.sh  
  
Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System  
$ ./fib.sh  
Enter the number of terms:  
5  
The Fibonacci series is:  
0 1 1 2 3  
  
Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System  
$ |
```

RESULT:

Thus, AWK scripting was successfully implemented to process text data, perform arithmetic calculations, and apply logical conditions to generate meaningful and formatted output.